ADAPTIVE ATHLETE

Combined Design + Technical PRD

MVP / Engineering Handoff • v1.0 • August 19, 2026

Training that adapts to the athlete's life without losing the training objective.

Field	Decision
Product stage	MVP definition and implementation handoff
Primary wedge	HYROX / hybrid racing; architecture must remain sport-extensible
Primary platforms	iOS first; Android parity planned; Apple Watch execution is a near-term extension
Core differentiator	Adaptive training queue + stimulus-preserving Full / Express / Micro workout variants
Recommended client	React Native + Expo + TypeScript, with native modules where HealthKit/WorkoutKit require them
Recommended backend	Supabase (PostgreSQL, Auth, Storage, Edge Functions)
AI role	Explain, classify, and propose adaptations within deterministic safety and programming constraints; never be the source of truth for workout structure

1. Executive Summary

Adaptive Athlete is a performance-training application for athletes whose schedules, recovery, equipment, and available time are not predictable. Instead of treating a calendar as the source of truth, the system maintains a goal-driven training queue and chooses the best valid session for the athlete's current state. The MVP launches with HYROX/hybrid training because the sport has a fixed race structure and measurable station demands, but the domain model must support future running, strength, and endurance programs.

The product's signature moment is adaptation: a planned 70-minute session can become a 30-minute Express or 15-minute Micro session while preserving the day's intended training stimulus. The athlete remains 'on track' when the correct reduced stimulus is completed; missed calendar dates do not create backlog.

MVP north star: The user opens the app, understands what matters today in under 10 seconds, adapts the session in under 30 seconds when needed, completes it with minimal logging friction, and sees the plan intelligently continue.

1.1 Goals

- Deliver a trustworthy daily recommendation based on program intent, completed training, recovery, time, equipment, and race proximity.
- Make Full, Express, and Micro sessions first-class successful outcomes rather than good/better/best labels.
- Preserve progressive overload and aerobic progression despite irregular training days.
- Provide transparent rationale for adaptations so athletes understand why the plan changed.
- Track performance trends and race readiness without centering weight loss.
- Create a workout/content architecture that coaches and internal operators can curate and QA.

1.2 Non-goals for MVP

- Open-ended AI generation of arbitrary workouts with no curated template constraints.
- Social feed, leaderboards, public profiles, clubs, or messaging.
- Nutrition/macronutrient coaching.
- Medical diagnosis, treatment, injury rehabilitation, or clinical postpartum care.
- Automatic race-time prediction presented as a guaranteed outcome.
- Coach marketplace or multi-athlete coach dashboard.
- Full Apple Watch standalone app in the first mobile release; data integration is MVP, dedicated watch execution may follow.

2. Product Principles

Principle	Requirement
Goal stays; path adapts	Adapt time, volume, modality, and sequence without casually changing the athlete's long-term objective.
Stimulus > calendar obedience	Completion is evaluated against required weekly stimuli and progression, not whether Tuesday's workout occurred Tuesday.
Deterministic core, AI-assisted edge	Rules select valid candidates; AI explains and handles natural-language context. AI cannot bypass contraindication or progression rules.
Low-friction execution	During workouts, large controls, one-hand use, auto-filled prior loads, minimal text entry.
Explain the why	Every adaptation must expose a short human-readable rationale and the stimulus being preserved.
Privacy by design	Health permissions are optional, granular, and requested only when a feature needs them.
Sport-extensible	HYROX concepts are implemented through sport/race definitions and station schemas, not hard-coded throughout the app.

3. Personas & Jobs To Be Done

3.1 Primary: Time-Constrained Competitive Athlete

A recreationally competitive athlete training 4-6 times per week when life allows. They understand RPE and basic training language, have a race or performance goal, and need help choosing the highest-value session when conditions change.

Jobs

- Tell me what to do today without making me rebuild my own program.
- If I only have 20-30 minutes, give me a legitimate session that keeps me progressing.
- If I slept badly or feel depleted, prevent one hard session from damaging the rest of the week.
- When equipment changes, preserve the intended stimulus with an appropriate substitution.
- Show me whether my running, strength, stations, and consistency are actually improving.

3.2 Secondary: Returning Athlete

An experienced athlete returning after pregnancy, injury, illness, or a long layoff. The system must distinguish 'deconditioned experienced athlete' from 'true beginner' and progress conservatively without erasing prior skill.

3.3 Future: Coach-Managed Athlete

Not MVP. The domain model should permit a coach-authored program, locked sessions, comments, and athlete overrides later.

4. HYROX Domain Requirements

The current HYROX singles format is eight 1 km runs interleaved with eight workout stations; the app should store the race definition as versioned data so station order, loads, divisions, and future rule changes are content updates rather than application releases.

Entity	Examples / Notes
Sport	HYROX, Running, Strength (future)
Race definition	Versioned station sequence, run segments, division rules
Division	Open, Pro, Doubles, Relay; configurable rather than enum-only
Station	SkiErg, sled push, sled pull, burpee broad jump, row, farmer carry, sandbag lunge, wall ball
Training quality	Aerobic base, threshold, race pace, strength, station durability, compromised running, recovery
Benchmark	1 km run, 1 mile, 1,000 m SkiErg, station tests, half simulation

5. Information Architecture & Navigation

Bottom navigation for MVP: Today, Plan, Progress, Coach, Profile.

Surface	Primary purpose	Key actions
Today	Decision surface	View recommendation; adapt; start workout; complete check-in
Plan	Training trajectory	View weekly stimuli, queue, phase, race roadmap; inspect upcoming sessions
Progress	Evidence of improvement	Readiness, running, SkiErg, strength, stations, consistency
Coach	Natural-language control layer	Ask why; report context; request adaptation/substitution; explain metrics
Profile	Configuration	Goals, races, equipment, units, health connections, privacy, notifications

6. Core User Flows

6.1 First-Time Onboarding

1. Account creation: Apple, Google, email magic link.
2. Select training goal/sport. MVP prioritizes HYROX.
3. Add race: event name, date, division intent, goal (finish healthy / performance / custom).
4. Experience profile: race history, training age, current run capacity, weekly training availability.
5. Equipment inventory: home/gym equipment with optional location profiles.
6. Constraints: typical session duration, schedule predictability, impact tolerance; optional return-to-training considerations.
7. Baseline metrics: optional body weight; working strength loads; current run pace/time; HR source.
8. Health permissions: defer until value is clear; explain exactly what is read/written.
9. Generate initial plan and show first-week stimulus queue.

6.2 Daily Recommendation

Step	System behavior
1. Open Today	Load active race, current phase, incomplete weekly stimuli, last 7-14 days training load, recovery state, equipment context.
2. Determine recommendation	Adaptation engine ranks eligible sessions and variants.
3. Present	Show one primary recommendation plus rationale; do not present a wall of choices.
4. User adapts if needed	Time, energy, low-impact, equipment, pain flag; recalculate.
5. Start	Create immutable workout_session snapshot from the selected template/variant.
6. Complete	Capture actuals, RPE, symptoms/notes, and optional HR/workout data.
7. Reconcile	Update stimulus completion, progression state, readiness inputs, and next queue order.

6.3 Adapt Workout

Inputs must be fast and mostly tap-based.

Input	MVP values	Effect
Available time	15 / 30 / 45 / 60 / 90+	Variant selection and block pruning
Energy	Low / Normal / High	Intensity ceiling and session ordering
Sleep/recovery	Automatic if connected; manual fallback	May downgrade high-intensity work
Low impact	Boolean	Removes/changes high-impact movements
Unavailable equipment	Multi-select	Applies validated substitutions or selects another session
Pain/symptom flag	Body area + severity + free text	Conservative adaptation; may recommend recovery/stop and advise appropriate professional evaluation rather than diagnose

6.4 Workout Execution

- Create workout session snapshot so later template edits do not change historical records.
- Support block types: warm-up, straight sets, supersets, rounds, AMRAP, EMOM, intervals, continuous, simulation, cooldown.
- Support prescriptions: reps, distance, duration, calories, load, RPE, pace, HR zone, rest.
- Auto-populate last successful load and allow quick +/- changes.
- Provide timer, round counter, rest timer, and large Complete/Skip controls.

- Allow exercise substitution only from validated substitution graph unless user explicitly chooses a free-form replacement.
- Persist locally during active workout and sync idempotently when connectivity returns.
- At completion collect session RPE, difficulty, notes, and optional symptom flags.

7. Functional Requirements

ID	Pri	Area	Requirement
FR-001	P0	Race management	User can create, edit, archive, and select an active race/goal.
FR-002	P0	Program generation	System creates a phase-based program with weekly required stimuli and a flexible session queue.
FR-003	P0	Daily recommendation	System returns exactly one recommended session/variant plus rationale.
FR-004	P0	Adaptive variants	Every eligible workout supports Full, Express, and Micro semantics or is explicitly marked non-compressible.
FR-005	P0	Time adaptation	Changing time budget recalculates the workout while preserving primary stimulus where possible.
FR-006	P0	Recovery adaptation	Low recovery can reduce intensity/volume or reorder the queue.
FR-007	P0	Equipment adaptation	System filters by available equipment and applies approved substitutions.
FR-008	P0	Workout player	User can execute structured blocks and log actual performance.
FR-009	P0	Strength logging	Load/reps/RPE are captured per set; prior loads are surfaced.
FR-010	P0	Cardio logging	Distance, duration, pace/split, HR and RPE can be recorded.
FR-011	P0	Offline active workout	Active workout remains usable with no network and syncs later.
FR-012	P0	Weekly stimulus completion	Plan shows completed/remaining stimuli independent of weekday.
FR-013	P0	Progression rules	Next exposure can change duration, load, reps, rounds or recovery based on deterministic rules.
FR-014	P0	Readiness	App computes transparent component scores; score is guidance, not a medical metric.
FR-015	P0	Recovery check-in	Manual sleep/energy/stress/soreness/motivation inputs available.
FR-016	P1	HealthKit read	With permission, ingest workouts, HR, resting HR, sleep where available, weight and relevant activity data.
FR-017	P1	HealthKit workout write	Write completed supported workouts where appropriate and permitted.
FR-018	P1	WorkoutKit	Export/schedule compatible structured workouts to Apple Workout/Watch where supported.
FR-019	P1	Health Connect	Android parity for exercise, HR, sleep, weight and related supported records.
FR-020	P0	AI Coach	Coach can explain, summarize, classify user context and request bounded adaptation actions.
FR-021	P0	AI action confirmation	Any AI-proposed program mutation is shown as a structured change before application when it changes more than today's session.
FR-022	P0	Content admin	Internal operators can version exercises, templates, substitutions, progression rules and race definitions.
FR-023	P1	Notifications	Adaptive, non-shaming reminders and race/phase prompts.
FR-024	P0	Data export/delete	User can request account deletion and export core training history.

8. Design Specification

8.1 Visual Direction

Premium athletic + technical. Dark graphite or warm-light neutral foundations, crisp typography, restrained visualization, and a single energetic accent. Avoid distressed CrossFit aesthetics and maternal-wellness visual cues.

Design tokens

Token group	Guidance
Typography	Modern grotesk/sans; tabular numerals for pace, HR, time and load; minimum body size 16sp equivalent on workout execution surfaces.
Spacing	8-point base grid; workout controls favor 16-24pt spacing.
Touch targets	Minimum 44x44 pt iOS / 48x48 dp Android; workout primary actions larger.
Variants	Consumer labels Full / Express / Micro. Do not communicate Express or Micro as failure states.
Semantic status	Ready/complete, caution/adapted, recovery concern, informational aerobic; never rely on color alone.
Charts	One metric per chart by default; explicit units; accessible contrast; no decorative chartjunk.

8.2 Screen Inventory

ID	Screen	Purpose
D01	Splash/Auth	Authentication and session restore
D02	Goal selection	Sport and primary outcome
D03	Race setup	Date/division/goal
D04	Experience baseline	History and current capacity
D05	Equipment	Equipment/location profiles
D06	Availability	Typical time and schedule predictability
D07	Health permissions	Progressive permission education
D08	Plan ready	Initial phase + first-week summary
D09	Today	Primary daily decision surface
D10	Adapt sheet	Time/recovery/equipment adaptation
D11	Workout detail	Purpose, blocks, targets, variants
D12	Active workout	Execution state machine
D13	Strength logger	Set-level load/reps/RPE
D14	Interval/round player	Timers, rounds, pace/HR
D15	Completion	Summary, RPE, notes, PRs
D16	Plan week	Stimulus completion + queue
D17	Race roadmap	Phases and race countdown
D18	Progress overview	Readiness components and trends
D19	Metric detail	Running/Ski/strength/stations
D20	Recovery check-in	Manual readiness inputs
D21	Coach	Conversation + structured actions
D22	Profile/settings	Units, privacy, notifications
D23	Equipment profiles	Home/gym/travel
D24	Connections	HealthKit/Health Connect state

8.3 Today Screen Acceptance Criteria

- Active race and days remaining visible above the fold.
- Primary recommendation name, purpose, estimated duration, and variant visible without scrolling on common phone sizes.
- Start Workout is the dominant CTA; Adapt is secondary but visually obvious.
- If data is insufficient, show a neutral 'baseline building' state instead of fabricated precision.

- If no workout is recommended because of a severe user-reported issue, show recovery/stop guidance and do not generate a hard session.

8.4 Active Workout State Machine

State	Allowed transitions	Persistence
ready	start, exit	session draft
active_block	pause, complete step, skip step, substitute, exit	local + periodic sync
rest	skip rest, auto advance, pause	local
paused	resume, end workout	local + server
completed_pending_review	save review, edit actuals	local + server
completed	none except notes/edit corrections	server canonical
abandoned	resume within grace period or finalize abandoned	server canonical

9. Adaptive Training Engine

The engine is a deterministic ranking and transformation service. It does not ask an LLM to invent a workout. It chooses among curated session intents/templates and validated transformations.

9.1 Inputs

- Active program phase and race date.
- Weekly required stimuli and completion status.
- Last completed sessions, volume, intensity and muscle/movement exposure.
- Progression state for running, SkiErg, strength and station skills.
- Recovery state: manual and/or connected data.
- Available time.
- Equipment/location profile.
- Impact restrictions and user-reported symptoms.
- User preferences: treadmill/outdoor, training days, variation tolerance.
- Hard constraints: minimum recovery between high-intensity or heavy lower-body exposures; taper rules.

9.2 Candidate Ranking

Recommended implementation: score eligible session intents, then select a compatible template and variant.

Component	Example weight	Description
Stimulus urgency	30%	How necessary this quality is to meet the week's/phase's training objectives.
Recovery fit	20%	Compatibility with current recovery and recent load.
Race specificity	15%	Increases as race approaches.
Progression continuity	15%	Preserves progression from prior exposure.
Time fit	10%	Can the session deliver meaningful stimulus within available time?
Equipment fit	5%	No invalid substitutions or missing required equipment.
Preference/adherence	5%	Variation and modality preference; never overrides safety/hard constraints.

9.3 Variant Transformation Rules

Transformation	Full → Express	Full → Micro
Warm-up	Shorten but retain movement-specific prep	Minimum safe prep
Primary quality	Preserve	Preserve one high-value dose when recovery

		allows
Accessory volume	Reduce first	Remove
Rounds/sets	~60-75% volume	~25-40% volume
Intensity	Usually preserve submaximal target	Do not increase intensity to compensate for lost time
Rest	May compress modestly if stimulus permits	Keep enough rest for quality
High impact	Retain only if user state allows	Prefer low-impact equivalent on poor-recovery days
Cooldown	Shorten	Optional brief reset

9.4 Hard Guardrails

- Never increase both intensity and volume as compensation for missed training.
- Never schedule 'make-up' double sessions solely because prior sessions were missed.
- Do not prescribe maximal testing during low-recovery state.
- Do not infer medical clearance. User-reported pain, pelvic pressure, leaking, dizziness, chest symptoms, or other concerning symptoms trigger conservative handling and appropriate safety messaging.
- Progression caps are stored by workout family and can be versioned.
- Taper phase rules override generic progression.
- If no valid session exists, return recovery/no-session rather than forcing a recommendation.

9.5 Readiness Model

Readiness is a product score, not a validated medical or physiological diagnosis. MVP should expose component scores and confidence.

Component	Suggested MVP calculation
Aerobic	Recent easy duration + threshold completion + comparable HR/pace trend
Running	Weekly run exposure + longest continuous run + 1 km/benchmark trend
Strength	Recent prescribed-load completion relative to baseline/goal requirements
Stations	Coverage and benchmark trend across race-specific stations
Consistency	Required stimuli completed over rolling 4 weeks, weighted by importance
Recovery	Manual recovery + connected sleep/HR signals when available

Overall score should be a weighted normalized composite with a confidence flag (low/medium/high). Do not show precise decimals. Product/coach team must validate weights against real athlete outcomes before marketing the score as predictive.

10. Workout Content Model

The existing seed database already follows the intended pattern: exercise primitives → equipment → workout templates → blocks → block exercises → Full/Express/Micro variants → tags/substitutions/progression rules. Production should retain this composable model.

Entity	Key fields
exercise	id, name, movement_pattern, modality, impact_level, default_unit, sport relevance, contraindication tags
equipment	id, name, category
exercise_equipment	exercise_id, equipment_id, required/optional
workout_template	family, primary/secondary goal, duration, intensity, impact, specificity, level range
workout_block	type, order, rounds/duration/rest, instructions
block_exercise	exercise, prescription type, quantity/unit, intensity target, side
workout_variant	Full/Express/Micro, time budget, volume multiplier, transformation metadata

substitution	source exercise, replacement, reason, priority, equivalence tags
progression_rule	family, metric, trigger, action, max change
race_definition	sport, season/version, division, ordered segments/stations

11. Data Model

Table	Core fields / responsibility
users	id, auth_id, locale, units, timezone, created_at
athlete_profiles	user_id, experience_level, training_age, preferences_json
races	id, user_id, race_definition_id, event_name, event_date, division, goal_type, goal_value, status
programs	id, user_id, race_id, version, start_date, end_date, status
program_phases	id, program_id, phase_type, start/end, objectives_json
weekly_cycles	id, phase_id, week_index, target_load, status
stimulus_requirements	id, weekly_cycle_id, stimulus_type, target_exposures, priority, min/max dose
session_queue_items	id, weekly_cycle_id, workout_intent_id, rank, state, earliest_at, expires_at
workout_templates	content entities from workout DB
workout_sessions	id, user_id, template_id, variant, snapshot_json, started_at, ended_at, status, session_rpe
session_blocks	session_id, snapshot of prescribed + actual block data
set_logs	session_block_id, exercise_id, set_index, prescribed/actual reps/load/rpe
cardio_logs	session_block_id, exercise_id, duration, distance, avg_hr, max_hr, pace/split, calories
recovery_checkins	user_id, date, sleep_score, energy, stress, soreness, motivation, symptom_json
health_daily_metrics	user_id, date, source, resting_hr, sleep_duration, weight, etc.
equipment_profiles	user_id, name, location_type
equipment_profile_items	profile_id, equipment_id, properties_json
adaptation_events	user_id, original_session, selected_session, inputs_json, reasons_json, engine_version
readiness_snapshots	user_id, date, overall, components_json, confidence, model_version
coach_threads/messages	conversation history and structured action payloads
content_versions	entity_type, entity_id, version, status, authored_by, published_at

11.1 Important Data Rules

- Historical workout sessions store a snapshot; template edits are never retroactive.
- All engine decisions store engine_version and reason codes for reproducibility.
- Health-derived values store source and observed timestamp; never silently overwrite manual values.
- Race definitions and workout content are versioned and publishable/draftable.
- Use soft delete/archive for program/content records referenced by history.
- Use UUIDs; server timestamps in UTC; preserve user's IANA timezone for local-day calculations.

12. API Contract

REST examples below are implementation-neutral; Supabase RPC/Edge Functions may implement the same contracts.

Method	Endpoint	Purpose	Response
POST	/v1/onboarding/plan	Create athlete baseline + initial program	Plan summary, phase, first queue
GET	/v1/today	Fetch today's decision payload	Race, readiness, recommendation, rationale
POST	/v1/adapt	Recalculate current recommendation	Selected template/variant + reason codes
POST	/v1/workout-sessions	Start immutable session	session_id + snapshot

PATCH	/v1/workout-sessions/{id}	Checkpoint active session	updated server revision
POST	/v1/workout-sessions/{id}/complete	Finalize session	summary + progression effects
GET	/v1/plan/week	Weekly stimuli and queue	requirements + queue items
GET	/v1/progress	Readiness and metric summaries	component series
POST	/v1/recovery-checkins	Manual check-in	updated recovery state
POST	/v1/coach/messages	Coach message	text + optional structured actions
POST	/v1/coach/actions/{id}/apply	Apply approved coach action	updated plan/session
POST	/v1/integrations/health/sync	Trigger/reconcile health sync	sync status

12.1 GET /v1/today response shape

```
{
  "date_local": "2026-08-19",
  "active_race": {"id":"...", "name":"Boston HYROX", "days_remaining":52},
  "phase": {"type":"build", "week":7, "total_weeks":16},
  "readiness": {"overall":74, "confidence":"medium", "components": {...}},
  "recommendation": {
    "queue_item_id":"...",
    "workout_template_id":"...",
    "variant":"full",
    "estimated_minutes":62,
    "primary_stimulus":"aerobic_durability",
    "reason_codes":["STIMULUS_DUE","RECOVERY_NORMAL","EQUIPMENT_MATCH"],
    "rationale":"Aerobic durability is the highest-priority incomplete stimulus this week."
  }
}
```

13. AI Coach Architecture

AI is a bounded orchestration and explanation layer. It receives structured athlete state and can call internal actions; it does not directly write arbitrary program rows.

13.1 Allowed AI capabilities

- Classify natural-language context into structured signals: time limit, energy, equipment, travel, soreness/pain flag, preference.
- Explain why a workout or progression was chosen using engine reason codes.
- Summarize recent training trends from structured metrics.
- Propose an adaptation by invoking the deterministic adaptation service.
- Suggest validated equipment substitutions.
- Draft a plan-change proposal (e.g., reduce this week's load) for user confirmation.

13.2 Disallowed / constrained

- No medical diagnosis or medical clearance.
- No invented race rules, exercise loads, or progression outside the content/rule service.
- No silent multi-week program mutation.
- No use of private health fields beyond granted purpose/context.
- No claiming readiness score is clinically validated or guarantees race performance.

14. Health & Wearable Integrations

iOS should use HealthKit for permissioned health/fitness data and can use WorkoutKit for compatible structured workout composition/scheduling to Apple Watch. Apple documents WorkoutKit support for custom interval, single-goal and pacer workouts, and HealthKit workout sessions for live workout data. Android should use Health Connect with granular permissions; Google currently documents Health Connect as the unified on-device health/fitness store and its 1.1.0 Jetpack client as stable.

14.1 MVP permission strategy

Data	Why	Default
Workout sessions	Import completed activity and reconcile plan	Ask when user connects health
Heart rate	Session intensity and trend analysis	Optional
Sleep	Recovery context	Optional; explain limitations/source variance
Weight	Optional trend only	Off by default unless user enables
Steps/activity	Context, not direct training credit by default	Optional
Write workout	Keep ecosystem history consistent	Separate explicit permission/setting

Never request broad health access during account creation. Request the minimum data types required for a feature, at the moment the user enables that feature.

15. Technical Architecture

Layer	Recommendation	Notes
Mobile	React Native + Expo + TypeScript	Use native Swift/Kotlin modules for HealthKit/WorkoutKit/Health Connect capabilities not covered by Expo modules.
State	TanStack Query + lightweight local UI store	Server state separate from active workout machine.
Active workout	Explicit state machine + durable local persistence	SQLite/local DB; checkpoint server opportunistically.
Backend	Supabase Postgres/Auth/Storage + Edge Functions	RLS on all user-owned data.
Engine	Versioned server-side TypeScript service	Pure deterministic functions where possible; unit-testable fixtures.
AI	OpenAI Responses API via server only	Structured schemas for classifier/actions; never expose API key client-side.
Analytics	Product analytics provider + server event mirror	Avoid sending sensitive free-text health notes to analytics.
Observability	Sentry-style crash/error + backend logs/traces	Include engine decision IDs, never raw sensitive notes in logs.
Content admin	Internal web console	Publish/version exercises, templates, rules, race definitions.

15.1 Offline Strategy

- Cache current week, workout templates referenced by the queue, exercise metadata, and last performance values.
- When a session starts, write session snapshot locally before first block begins.
- Every completion event uses a client-generated event UUID and monotonically increasing session revision.
- Sync API is idempotent; duplicate events do not duplicate sets/metrics.

- If server plan changes while a workout is active, active snapshot wins; reconcile after completion.

15.2 Security

- Supabase Row Level Security: user can only read/write own athlete data; published workout content read-only.
- Service-role credentials only in server environment.
- Encrypt transport with TLS; rely on managed at-rest encryption plus platform secure storage for tokens.
- Store refresh/session tokens in Keychain/Keystore-compatible secure storage.
- Separate operational analytics identifiers from health data; minimize event payloads.
- Audit log for admin content publishing and engine/rule changes.
- Account deletion job removes user-owned records and disconnects integrations subject to legal retention requirements.

16. Analytics & Event Taxonomy

Event	Properties / notes
onboarding_started	source
onboarding_completed	sport, race_added, health_connected
today_viewed	phase, recommendation_family, variant
adapt_opened	original_variant
adapt_input_changed	time_bucket, energy_bucket, low_impact, equipment_change
adaptation_applied	from_variant, to_variant, reason_codes
workout_started	template_id, family, variant, estimated_minutes
workout_step_completed	block_type, exercise_id; sample/aggregate rather than every tap if volume is excessive
exercise_substituted	from_exercise, to_exercise, reason
workout_completed	family, variant, actual_minutes, session_rpe
workout_abandoned	elapsed_minutes, block_index
recovery_checkin_completed	buckets only
readiness_viewed	confidence, component_lowest
coach_message_sent	intent classification only; do not mirror raw sensitive text
coach_action_applied	action_type
health_connection_started/completed/failed	provider, permission_group

16.1 Product KPIs

Metric	Definition
Activation	User creates goal + completes first workout within 72 hours
Week-2 retained athlete	Completes ≥2 required stimuli in second program week
Adaptation utilization	% active users applying Express/Micro or other adaptation at least once/week
Adaptive save rate	% adapted workouts completed after adaptation
Stimulus adherence	Weighted required stimuli completed / required stimuli prescribed
Workout completion	Started sessions completed
Decision latency	Median time from Today open to workout start/adapt choice
Trust proxy	% recommendations accepted without manual replacement + low reversal rate

17. Notifications

Notifications are supportive and adaptive, never punitive.

Trigger	Example behavior
Planned training window	'Your next priority is aerobic durability. Full, Express, or Micro all count today.'

No training 48h + incomplete high-priority stimulus	Offer a short valid option, not a missed-workout warning.
Race phase transition	Explain what changes in the next phase.
Recovery low	If opted in, suggest opening Today to adapt rather than sending alarming language.
PR / benchmark	Celebrate measured improvement with restrained language.

18. Accessibility

- WCAG 2.2 AA target for mobile surfaces where applicable.
- Dynamic Type / font scaling without clipping primary workout controls.
- VoiceOver/TalkBack labels for timers, set controls, charts and readiness components.
- Do not encode workout variant, readiness or completion by color alone.
- Haptics optional; all haptic information duplicated visually.
- Charts provide text summaries and accessible values.
- Workout execution supports reduced motion.

19. Content Operations & QA

19.1 Publishing workflow

draft → coaching review → QA validation → published → deprecated. Published content is immutable by version; edits create a new version.

19.2 Required automated validations

- All exercise references exist and have compatible units.
- All required equipment is resolvable or template is explicitly non-adaptable.
- Express/Micro variants preserve a declared primary stimulus.
- No variant exceeds its time bucket using estimated block duration.
- Progression rules have max-change caps.
- High-impact exercises have impact tags and low-impact substitution path where applicable.
- HYROX race definitions pass station-order and distance validation for their rulebook version.

20. Edge Cases

Scenario	Expected behavior
User misses 10 days	Do not dump backlog. Reassess current capacity/recovery, preserve phase objective, and insert a bridge/re-entry sequence if thresholds are met.
Race date changes	Recalculate phase boundaries; preserve completed history; create new program version.
User switches Open ↔ Doubles	Update race requirements and specificity; do not erase base training.
No health permissions	All core functionality remains available with manual RPE/check-ins.
Health data conflict	Show source priority; do not average incompatible duplicates blindly.
Only 12 minutes available	Select Micro if valid; otherwise return recovery/mobility rather than fake the full stimulus.
Equipment unavailable mid-workout	Offer approved substitution; snapshot the change.
Poor connectivity	Continue active workout locally.
User reports severe symptoms	Stop adaptation flow for hard training and show conservative safety guidance; do not diagnose.

Readiness confidence low	Display 'building baseline' and component data rather than over-precise score.

21. MVP Release Scope

21.1 P0 - Launch

- Auth, onboarding, athlete profile, race/goal management.
- HYROX program phases, weekly stimuli, flexible queue.
- Curated workout database and Full/Express/Micro variants.
- Deterministic recommendation/adaptation engine.
- Today, Plan, workout detail/player, completion, Progress, Coach, Profile.
- Manual recovery and performance logging.
- Running, SkiErg, strength, station metrics.
- Readiness v1 with confidence.
- Offline active workout and sync.
- Content admin/versioning.
- Account export/delete and core privacy controls.

21.2 P1 - Immediately after launch / beta dependent

- HealthKit ingestion and workout write.
- WorkoutKit scheduling/export where compatible.
- Health Connect Android parity.
- Push notification personalization.
- More advanced race simulations and post-race report.
- December/next-race chaining and recovery blocks.

21.3 P2

- Dedicated Apple Watch workout app.
- Garmin/WHOOP/Oura/Strava direct integrations where valuable.
- Coach dashboard and coach-authored programs.
- Additional sports.
- Community/social features.
- Video form analysis.

22. Engineering Epics & Suggested Build Order

Epic	Name	Deliverable
E1	Foundation	Repo, CI/CD, auth, environment config, design system, navigation, error/analytics plumbing
E2	Content domain	Exercises, equipment, templates, variants, substitutions, admin seed/import
E3	Athlete + race	Profile, onboarding, race definitions, race CRUD
E4	Program service	Phases, weekly cycles, stimuli, queue, program versioning
E5	Adaptive engine	Eligibility, ranking, transformation, reason codes, fixtures/tests
E6	Today + Adapt	Daily payload, Today UI, adaptation sheet, rationale
E7	Workout execution	Snapshot, state machine, set/cardio logs, timers, offline persistence
E8	Reconciliation	Completion, progression effects, queue updates, sync/idempotency
E9	Progress	Metric aggregation, readiness v1, charts, confidence
E10	Recovery	Manual check-in and recovery state
E11	AI Coach	Structured classifier, explanations, bounded actions, confirmation flow

E12	Integrations	HealthKit/WorkoutKit; Android Health Connect in parity track
E13	Notifications	Permission, scheduling, adaptive copy
E14	Privacy/admin	Export/delete, audit logs, content publishing, support tools

23. Test Strategy

23.1 Unit tests

- Engine ranking with deterministic fixtures.
- Progression caps and deload/taper overrides.
- Variant transformation preserves primary stimulus.
- Equipment substitution graph.
- Readiness component normalization/confidence.
- Timezone/week-boundary calculations.

23.2 Integration tests

- Onboarding → plan → Today → adapt → workout → completion → next recommendation.
- Offline session → reconnect → idempotent sync.
- Template version update does not mutate historical session.
- Race date/division change creates correct program version.
- Health permission denied/revoked paths.
- AI action calls deterministic service and requires confirmation for multi-day mutations.

23.3 E2E acceptance scenarios

- User with 60 minutes and normal recovery receives Full workout and completes it.
- Same athlete with 25 minutes receives Express/Micro valid adaptation with rationale.
- Low-recovery athlete scheduled for threshold work receives reduced/reordered training according to rules.
- Missing SkiErg selects approved equivalent or different session without breaking weekly objective.
- User misses 7+ days and receives re-entry logic rather than backlog.
- User can complete a workout with network disabled and later see correct progress.

24. Definition of Done for MVP

- All P0 functional requirements pass acceptance tests.
- No critical/high security findings.
- Crash-free workout execution meets release target in beta.
- Engine decisions are reproducible from stored inputs + engine version.
- Every published workout has coaching review and valid variants/substitutions.
- Core flows usable without any health permissions.
- Accessibility audit passes agreed AA criteria for P0 screens.
- Privacy policy, consent copy, data export and deletion are implemented.
- Beta users can complete four weeks of irregular training without manual database intervention.

25. Open Product Decisions

Decision	Recommendation for MVP
Final brand name	Keep code/product working name Adaptive Athlete until trademark/domain screen is

	complete.
Readiness weights	Ship as configurable server-side model; validate during beta.
Calendar vs queue	Queue is primary; calendar is secondary visualization.
Green/Yellow/Red naming	Use Full / Express / Micro in consumer UI.
AI plan mutation	Today's adaptation may be one-tap; multi-day changes require confirmation.
Apple Watch	Use HealthKit/WorkoutKit integration first; standalone watch experience P2 unless beta proves execution demand.
Race data source	Maintain versioned internal race definitions with rulebook provenance; do not scrape at runtime.

26. External Platform Notes / Engineering References

The following platform facts were verified against current official documentation during PRD preparation:

- HYROX singles uses eight 1 km runs interleaved with eight workout stations; store this as versioned race content rather than hard-coded UI logic.
- Apple HealthKit is the permissioned repository for health and fitness data on iPhone/Apple Watch.
- Apple WorkoutKit supports creating/scheduling structured workouts, including custom interval workouts, to the Workout app on Apple Watch.
- Apple HealthKit workout APIs support live workout sessions and workout data collection.
- Android Health Connect provides permissioned read/write access to health and fitness records; the Jetpack Health Connect 1.1.0 release is stable as of August 2026.

Appendix A. Reason Codes

Code	Meaning
STIMULUS_DUE	Required weekly stimulus is incomplete/high priority
RECOVERY_HIGH/NORMAL/LOW	Recovery compatibility
TIME_LIMIT	Selected variant because of available time
EQUIPMENT_MATCH	All required equipment available
EQUIPMENT_SUBSTITUTION	Validated replacement used
IMPACT_REDUCTION	Low-impact preference/constraint applied
PROGRESSION_CONTINUITY	Maintains next step in progression
RACE_SPECIFICITY	Prioritized due to proximity to race
RECENT_LOWER_LOAD	Avoids conflicting lower-body stress
TAPER_OVERRIDE	Race taper rule applied
REENTRY_AFTER_GAP	Bridge logic after training gap
NO_VALID_HARD_SESSION	Recovery/no-session returned due to constraints

Appendix B. Suggested Repository Structure

```
apps/
  mobile/
  admin/
packages/
  domain/
  workout-engine/
  content-schema/
  ui/
  analytics/
supabase/
```

```
  migrations/
  functions/
    today/
    adapt/
    complete-workout/
    coach/
tests/
  engine-fixtures/
  e2e/
```

Appendix C. Handoff Artifacts

Engineering should receive this PRD together with:

- Workout seed database / schema (SQLite + SQL + JSON) already created for the project.
- Figma design system and P0 screen flows mapped to D01-D24.
- OpenAPI or typed contract generated from Section 12.
- Engine fixture pack covering normal, time-limited, low-recovery, equipment-limited, re-entry and taper scenarios.
- Race-definition seed data with provenance/version.
- Analytics tracking plan derived from Section 16.